Advance Python for Data Analysis

(APDA)

Lecture 1 - Introduction to Data Analysis and the Python Ecosystem

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Objective is to learn how to take raw data and extract useful information from it

Imagine you own a supermarket.

You have data for **100,000 transactions**.

What would you like to know?

- Which products sell the most?
- When are sales highest?
- Which products are frequently purchased together?
- Which customers spend the most?
- Why did sales decrease last month?
- What will sales be next month?

A dataset itself is not useful unless we can extract meaning from it

Data Is Everywhere

- Every activity generates data

E-commerce

`Customer | Product | Price | Date | Rating`

Healthcare

`Patient | Age | BP | Glucose | Diagnosis`

Agriculture

`Date | Rainfall | Soil Moisture | Temperature | Yield`

Finance

`Date | Stock Price | Volume | Return`

Social Media

`User | Post | Likes | Comments | Shares`

Sensors

`Timestamp | Temperature | Humidity | Pressure`

- Different domains generate different data, but the fundamental analysis process is surprisingly similar.
- Collect → Clean → Explore → Analyze → Interpret → Communicate

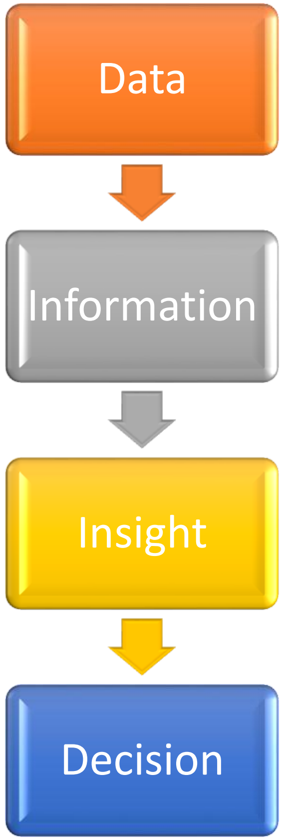
What is Data?

- Raw facts, measurements, observations, or records collected about something.

	A	B	C
1	number_courses	time_study	Marks
2	3	4.508	19.202
3	4	0.096	7.734
4	4	3.133	13.811
5	6	7.909	53.018
6	8	7.811	55.299
7	6	3.211	17.822
8	3	6.063	29.889
9	5	3.413	17.264
10	4	4.41	20.348

Information vs Data?

- Only marks column alone is nothing.
- Student and marks together makes meaningful information.



- Data 100, 120, 90, 75, 60 is just a number
- Information: Sales are decreasing.
- Insight: Sales decrease mainly occurs on weekends
- Decision: Introduce a weekend promotion

- The purpose of data analysis is not simply to calculate numbers. It is to support understanding and decisions

WHAT IS DATA ANALYSIS?

- Data Analysis is the process of inspecting, cleaning, transforming, exploring, and interpreting data to discover useful information, identify patterns, answer questions, and support decision-making

Raw Data

Inspecting

What data do we have?

Cleaning

Is something missing or incorrect?

Transforming

Can we convert the data into a useful format?

Exploring

What patterns exist?

Analyzing

Can we quantify relationships?

Interpreting

What does the result actually mean?

Student	Hours Studied	Attendance	Marks
A	2	60%	45
B	4	75%	62
C	6	90%	82
D	7	95%	91
E	3	70%	55

What can we learn from this tiny dataset?

- Average marks?
- Highest-performing student?
- Does studying more relate to higher marks?
- Does attendance affect performance?
- Can we predict marks for a new student?

Why Data Analysis?

Examples:

1. Netflix

Which movie should you watch?

2. Amazon

Which products should be recommended?

3. Google Maps

Fastest Route

4. Spotify

Recommended Songs

5. Weather

Rain Prediction

Four Important Types of Data Analysis

```
DESCRIPTIVE
What happened?

DIAGNOSTIC
Why did it happen?

PREDICTIVE
What is likely to happen?

PRESCRIPTIVE
What should we do?
```

Example:

Sales dropped from:

```
₹10 lakh → ₹7 lakh
```

Descriptive:

> Sales decreased by 30%.

Diagnostic:

> Why?

Predictive:

> Will the decline continue?

Prescriptive:

> What action should management take?

TYPES OF DATA: Structured, Semi-Structured Data, Unstructured Data

- Data Is Not All the Same

1. Structured Data: Data organized into clearly defined **rows and columns**

Example:

ID	Name	Age	Salary
101	Rahul	22	35000
102	Priya	24	42000
103	Aman	21	30000

Sources:

CSV

Excel

SQL databases

Pandas NumPy SQL

2. Semi-Structured Data: Data does not follow a traditional table but still contains organizational structure

JSON

```json
{
  "name": "Rahul",
  "age": 22,
  "skills": ["Python", "SQL"]
}
```

Examples:

- JSON
- XML
- Web APIs
- Log files

3. Unstructured Data

Examples:

- Images
- Videos
- Audio
- Emails
- PDFs
- Social media posts
- Documents

Images → Computer Vision

Text → NLP / LLMs

Another Classification: Numerical vs Categorical

```
DATA

├── Numerical
│       ├── Discrete
│       └── Continuous
│
└── Categorical
        ├── Nominal
        └── Ordinal
```

```
Age = 25                        → Numerical
Temperature = 32.5°C            → Numerical
Number of students = 60         → Discrete

City = Delhi                    → Categorical
Blood Group = B+                → Nominal
Rating = Poor/Good/Best         → Ordinal
```

DATA ANALYSIS WORKFLOW

```
1. Define the Problem

        ↓

2. Collect Data

        ↓

3. Understand Data

        ↓

4. Clean Data

        ↓

5. Explore Data

        ↓

6. Analyze / Model

        ↓

7. Visualize & Communicate

        ↓

8. Make Decisions
```

- Most beginners think data analysis starts with writing Python code. It doesn't."
- It starts with: **What question are we trying to answer?**

Step 1: Define the Problem

Example:

Bad problem:

> "Analyze this sales dataset."

Better:

> "Why did sales decrease in June?"

Even better:

> "Which products and regions contributed most to the June sales decline?"

The quality of analysis depends heavily on the quality of the question.

Step 2 & 3: Collect and Understand

```
CSV
Excel
Database
API
Sensors
Websites
Surveys
```

Ask these questions

```
How many rows?

How many columns?

What does each column mean?

What are the units?

Are values missing?

Are there duplicates?

Are data types correct?
```

Real Data Is Messy

Name	Age	Salary	City
Rahul	25	50000	Delhi
Priya	?	62000	Mumbai
rahul	25	50000	DELHI
Aman	-5	45000	Delhi
Neha	27	₹55,000	

What problems can you identify?

Missing age
Duplicate person?
Negative age
Inconsistent capitalization
Missing city
Salary stored with currency symbol

Exploratory Data Analysis: EDA

EDA = Getting to Know Your Data

building a machine learning model

Questions:

What is the distribution?

What is the average?

Are there outliers?

Are variables related?

Are there unusual patterns?

Are there missing values?

Why Python for Data Analysis?

```
                PYTHON
                   │
    ┌──────────────┼──────────────┐
    ↓              ↓              ↓
  NumPy          Pandas       Matplotlib
Numerical    Data Analysis  Visualization

    ↓              ↓              ↓
  SciPy          Seaborn        Plotly
Scientific     Statistical   Interactive
Computing      Visualization  Visualization

                   ↓
             Scikit-learn
           Machine Learning

                   ↓
          PyTorch / TensorFlow
             Deep Learning
```

If you had one million rows

Excel	Python
Slow	Fast
Manual	Automatic
Limited	Powerful
Small Data	Big Data
Few Charts	Advanced Visualization

python

INTRODUCTION

- Python was created by Guido van Rossum, and released in 1991.
- Python is the most widely used language for AI/ML because of its simplicity and powerful libraries.
- It allows fast prototyping and is used across both industry and research.
- Python is a high level language.
- Python is a case sensitive language. Ex: print() function

1. Python makes you job-ready faster.
2. Python helps you enter AI/ML, the fastest-growing field in the world.
3. Python is beginner-friendly yet powerful for experts.
4. Python is used by top companies in real AI systems.
5. Python has the best ecosystem for learning data science and machine learning.

Python is the gateway skill to Machine Learning, Deep Learning, NLP, Computer Vision, and Generative AI (GenAI).

1. Python for ML & DL:

- Python is high-level → Focus on logic, not boilerplate
- Example: C++ Matrix Multiply vs Python (NumPy):

```cpp
for(int i=0; i<n; i++){
    for(int j=0; j<m; j++){
        C[i][j] = A[i][j] * B[i][j];
    }
}
```

```python
C = A * B
```

- Python lets you think like a data scientist, not like a compiler.

2. Python for NLP (LLMs)

- Text preprocessing is super easy in Python
- Python runs tokenization, model inference and returns whether a review is positive or negative - used in real-world apps like automated customer-feedback analysis.

```python
from transformers import pipeline
nlp = pipeline("sentiment-analysis")
print(nlp("This product exceeded my expectations!"))
# -> [{'label': 'POSITIVE', 'score': 0.9998}]
```

Text → Preprocessing → Tokenizer → Embeddings → Transformer Model → Output

3. Python for Computer Vision (CV)

Python is used in CV because:

- OpenCV (Python API)
- PyTorch vision models
- CNN, ResNet, YOLO, ViT

```python
import cv2

img = cv2.imread("image.jpg")
edges = cv2.Canny(img, 100, 200)

cv2.imshow("Edges", edges)
cv2.waitKey(0)
```

Python reads an image, performs edge detection, and displays the result — a common step in computer vision tasks like object detection or feature extraction.

4. Python for GenAI + VLM (Vision-Language Models)

Python is used for:

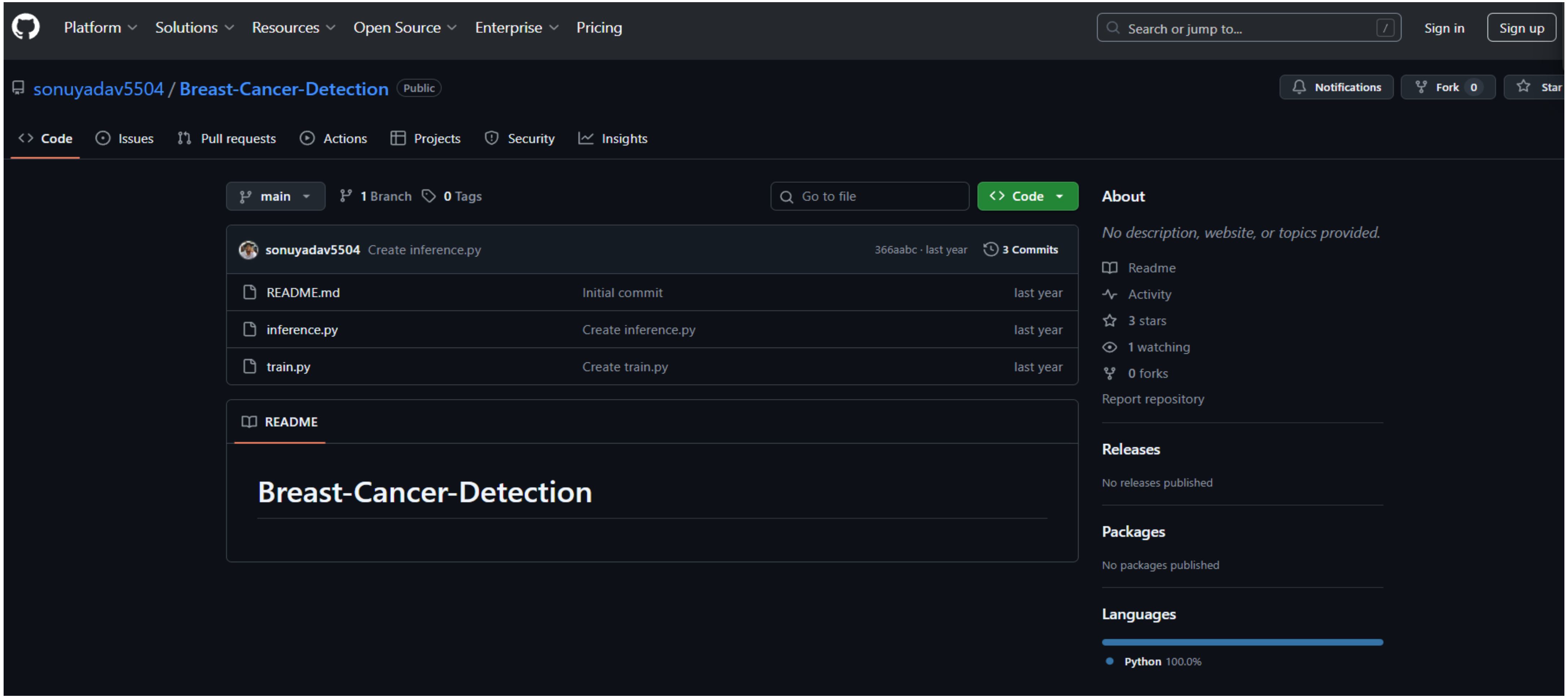
- Diffusion models (Stable Diffusion)
- VLMs like CLIP, Flamingo, Llava
- Image + Text multimodal models

```
from diffusers import StableDiffusionPipeline
pipe = StableDiffusionPipeline.from_pretrained("runwayml/stable-diffusion-v1-5")
img = pipe("a futuristic robot teaching Python").images[0]
```

Text Prompt → Model → Generated Image

Role of Python in Research

- 90% research papers in ML/CV/NLP
- Kaggle competitions
- AI academic labs



https://github.com/sonuyadav5504/Breast-Cancer-Detection

The Libraries You MUST Know

These are the tools we will master:

1. NumPy

- Foundation of numerical computing
- Behind the scenes of ML/DL math

```
import numpy as np
arr = np.array([1,2,3])
print(arr * 2)
```

Output: [2 4 6]

2. Pandas

- Used for cleaning, transforming, analyzing datasets

```
import pandas as pd
df = pd.read_csv("data.csv")
df.head()
```

3. Matplotlib / Seaborn

- Used for visualization, EDA, research plots

```
import matplotlib.pyplot as plt
plt.plot([1,2,3],[2,4,1])
plt.show()
```

4. Scikit-Learn

- Classical ML algorithms (Linear Regression → Random Forest → Clustering)

```
from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X, y)
```

5. PyTorch

- Deep learning backbone
- Build neural networks, CNN, LSTM, Transformers

```
import torch
x = torch.tensor([2.0], requires_grad=True)
y = x**2
y.backward()
x.grad
```

How to run Python code?

- There are many ways to run python code

CATEGORY 1 — TERMINAL / COMMAND-LINE ENVIRONMENTS

These run Python directly from the system shell.

1. Command Prompt (Windows)
2. Terminal (Mac/Linux)
3. Python REPL (opened by typing python)
4. PowerShell
5. Google/Microsoft online Python consoles (like Python.org interpreter or MS Learn sandbox)

Use case: For quick testing, running scripts, automation, backend jobs.

To run code: ->

-> First download python

-> python –version

-> python file.py

CATEGORY 2 — SCRIPT EDITORS & IDEs (For Writing .py files)

These are apps where you write full Python programs.

2A. **Text Editors** (basic, not intelligent)

- Sublime Text
- Atom
- Notepad

They only show syntax; no debugging, no auto-complete.

2B. **Full IDEs (Integrated Development Environments)**

These include project tools, debugging, autocomplete, and integration.

- VS Code (technically a lightweight IDE, but most popular)
- PyCharm
- Spyder
- Anaconda Navigator (launcher for Spyder, Jupyter, etc.)

Use case: Real projects, ML codebases, debugging, working with multiple files.

CATEGORY 3 — NOTEBOOK ENVIRONMENTS (Cell-based, Interactive)

- Jupyter Notebook
- JupyterLab
- Google Colab
- Kaggle Notebooks

CATEGORY 4 — CLOUD PYTHON ENVIRONMENTS (Online, No Install)

You run Python completely in the cloud.

- Google Colab
- Kaggle Notebook
- Replit
- Binder
- AWS SageMaker Studio
- Azure ML Notebooks

Many of these use Jupyter Notebook behind the scenes, but hosted online.

Print(“Hello World”)